

# Chi-square Tests

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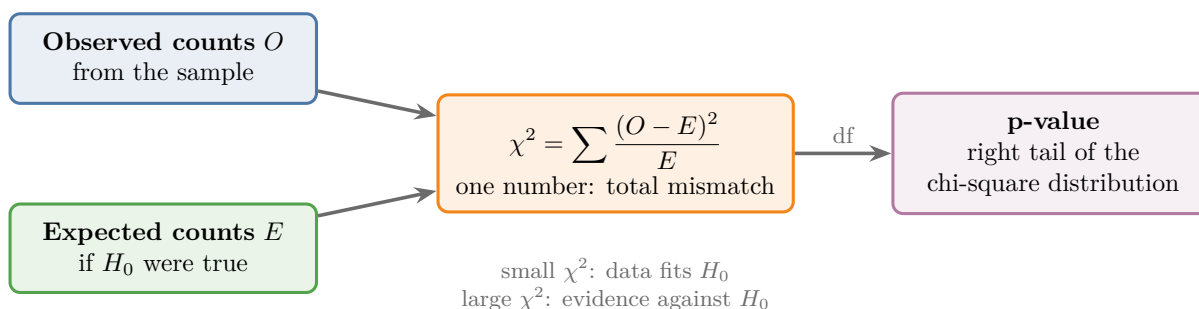
Where this fits: Pipeline map step 0 (Foundations). See the [Course map](#).



- **Builds on:** Hypothesis testing: null and alternative ([Note MA-038](#)); Joint and marginal probability ([Note MA-014](#)).
- **Compare with:** One-sample proportion test ([Note MA-044](#)).

## 1. Overview

**Key point:** A chi-square test compares the counts we observe in each category with the counts we would expect if  $H_0$  were true; the bigger the total mismatch  $\chi^2$ , the smaller the p-value.



**Figure 1:** The chi-square test in four steps

The chi-square test was named in the [what is statistics Note](#) as the test for categorical data. In the [choosing a hypothesis test Note](#) the chi-square test is the test for two categorical **features** (G-772; a feature is a variable of the data, one column of the data table, such as gender). Figure 1 shows how every chi-square test works: observed counts and expected counts go into one number,  $\chi^2$ , and its tail area under the chi-square distribution is the p-value.

This Note covers:

- the chi-square statistic and the chi-square distribution;
- the **goodness-of-fit test**: does one categorical feature match claimed proportions?
- the **test of independence**: are two categorical features related?
- a case study on the Titanic, the conditions the test needs, and its strength measure, Cramér's V.

**Extra:** All of this Note is extra material beyond the “which test when” guide; the guide only names the chi-square test.

## 2. The chi-square statistic

**Key point:**  $\chi^2 = \sum(O - E)^2/E$  adds up, over all cells, the squared gap between observed and expected counts, each scaled by the expected count.

### 2.1 Observed and expected counts

**Key point:** Observed counts come from the data; expected counts are what  $H_0$  predicts for a sample of the same size.

A categorical feature is summarised by counts, one per category (see the [frequency tables Note](#)). For a chi-square test we need two sets of counts:

- **observed counts**  $O$  (G-1375): how many **observations** (G-1374; records, one row of the data table each) of the sample fall in each category or cell;
- **expected counts**  $E$  (G-723): how many would fall there, on average, if  $H_0$  were true.

If  $H_0$  is true,  $O$  and  $E$  differ only by chance. A large mismatch is evidence against  $H_0$ .

### 2.2 The formula

**Key point:** Square each gap so that positive and negative gaps both count, divide by  $E$  so that a gap of 5 matters more in a small cell than in a big one, and add.

1. **In words:** for each cell, take the gap between observed and expected, square it, divide by expected; add over all cells.
2. **Formula:**

$$\chi^2 = \sum_{\text{cells}} \frac{(O - E)^2}{E}$$

3. **Example:** three cells with  $O = 20, 26, 14$  and  $E = 15, 33, 12$  (section 4):

$$\frac{(20 - 15)^2}{15} = 1.667$$

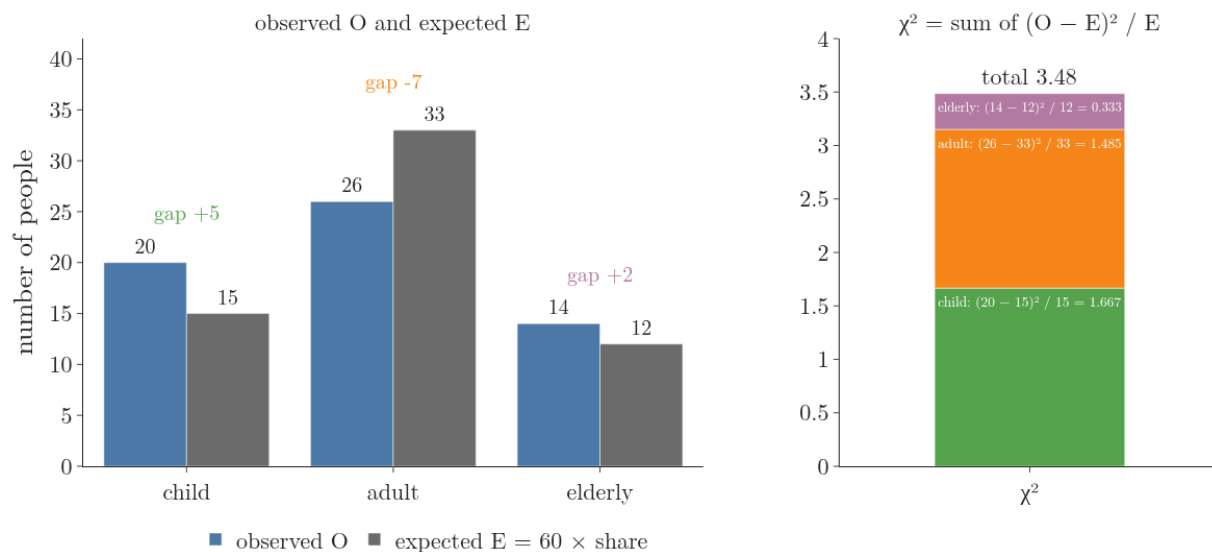
$$\frac{(26 - 33)^2}{33} = 1.485$$

$$\frac{(14 - 12)^2}{12} = 0.333$$

$$\chi^2 = 1.667 + 1.485 + 0.333 = 3.48$$

$\chi^2$  is 0 only when every observed count equals its expected count.  $\chi^2$  can never be negative. This number is the **chi-square statistic** (G-379).

Figure 2 builds it for the example. Each category has its observed and expected bar; the gaps are +5, -7 and +2. Squaring and dividing by  $E$  turns each gap into one block, and the blocks stack into  $\chi^2 = 3.48$ . The adult gap is the largest, but its block is smaller than the child block, because 7 out of an expected 33 is a smaller surprise than 5 out of an expected 15.



**Figure 2:** The chi-square statistic built cell by cell for the age groups against the census shares. Left: observed and expected counts, with each gap. Right: each cell's  $(O - E)^2 / E$  stacked into the total, 3.48.

### 3. The chi-square distribution

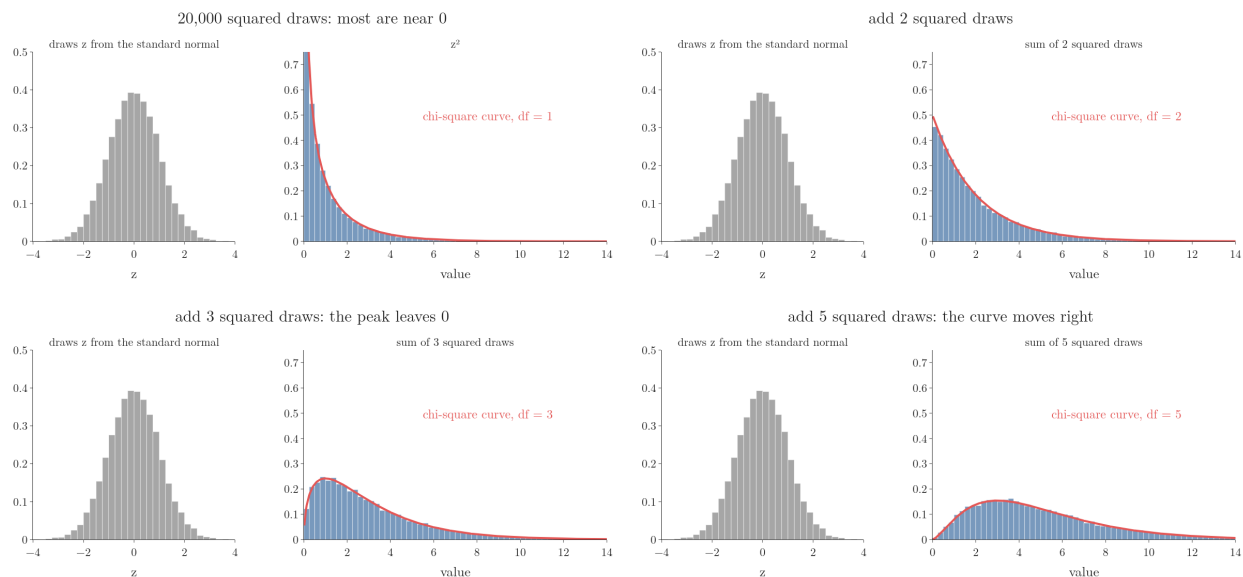
**Key point:** If  $H_0$  is true,  $\chi^2$  follows a chi-square distribution: never negative, skewed to the right, with one parameter, the degrees of freedom.

#### 3.1 Built from squared normal draws

**Key point:** Square a draw from the standard normal distribution and we get a chi-square value with 1 degree of freedom; add  $k$  squared draws and we get  $k$  degrees of freedom.

To judge whether  $\chi^2 = 3.48$  is large, we need to know which values  $\chi^2$  takes by chance. The curve that answers this is built from the standard normal distribution (see the [standard normal Note](#)) in two steps. Figure 3 runs them.

1. **Square one draw.** Take a number  $z$  from the standard normal distribution and square it. Most draws lie between  $-1$  and  $1$ , and squaring a number below  $1$  makes it smaller still, so most squares pile up next to  $0$ . A square of  $4$  needs a draw of  $2$  or  $-2$ , which is rare. A square is never negative.
2. **Add several squares.** Add the squares of  $2$  independent draws, then of  $3$ , then of  $5$ . The sum is rarely near  $0$  now, because all the draws would have to be small at once. The peak leaves  $0$  and the whole curve moves to the right.

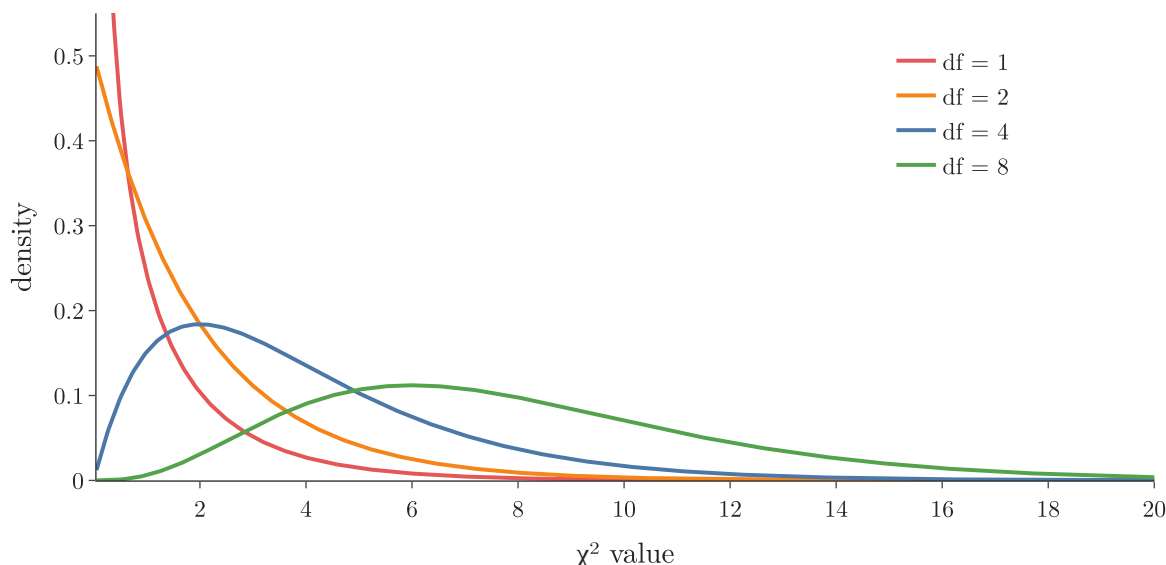


**Figure 3:** Left: 20,000 draws from the standard normal distribution. Right: the histogram of one squared draw, then of the sum of 2, 3 and 5 squared draws, with the chi-square density of 1, 2, 3 and 5 degrees of freedom (red). Idea after Khan Academy, “Chi-square distribution introduction”.

The distribution of a sum of  $k$  squared standard normal draws is the **chi-square distribution** (G-378), and  $k$  is its **degrees of freedom** (G-578; df). This is its definition.

### 3.2 The curves and the tail area

**Key point:** More degrees of freedom move the curve to the right; the p-value is the area to the right of our  $\chi^2$ .



**Figure 4:** Chi-square densities for four degrees of freedom

The chi-square distribution is a continuous distribution on the positive numbers. Like Student's t-distribution (see the [t-procedure Note](#)), it has one parameter, the degrees of freedom. Figure 4 shows four of them:

- with 1 or 2 df, the density is highest at 0 and falls steadily;
- with more df, the peak moves right and the curve spreads out;
- the mean of a chi-square distribution equals its df (NIST Handbook §1.3.6.6.6).

Only large  $\chi^2$  values count as evidence against  $H_0$ , so the p-value is always the **right-tail** area beyond our  $\chi^2$  (a **right-tailed test**, G-1693). The 5% **critical values** (G-504) are 3.84 for 1 df, 5.99 for 2 df and 7.81 for 3 df.

**Extra:** Why the statistic of section 2 follows this curve. Each  $(O - E)/\sqrt{E}$  is roughly normal for large counts, so  $\chi^2$  is a sum of squared, roughly normal terms. The terms are not independent, because the counts must add up to  $n$ . Pearson (1900) showed that  $\chi^2$  still follows a chi-square distribution for large samples, with one df lost to that constraint. The notebook checks this: 200,000 samples of 60 people drawn with the census shares of section 4 true give a mean  $\chi^2$  of 2.00, equal to  $df = 2$ , and 4.9% of them pass the 5% critical value 5.99.

#### 4. The goodness-of-fit test

**Key point:** One categorical feature against claimed proportions: expected count = sample size  $\times$  claimed proportion, and the degrees of freedom are the number of categories minus 1.

## 4.1 The question

**Key point:** Do the age groups in our sample of 60 match a population that is 25% children, 55% adults and 20% elderly?

The **goodness-of-fit test** (G-853) asks whether the counts of one categorical feature fit a claimed distribution. We use the 60 people of the [choosing a hypothesis test Note](#). Suppose a census says the population is 25% children, 55% adults and 20% elderly.

- $H_0$ : the age groups occur in the claimed proportions 0.25, 0.55, 0.20;
- $H_1$ : at least one proportion differs.

## 4.2 Expected counts and the statistic

**Key point:** each expected count is 60 times the claimed share: 15 children, 33 adults and 12 elderly. Together they give  $\chi^2 = 3.48$ .

1. **In words:** the expected count of each category is the sample size times its claimed share.
2. **Formula:**

$$E_i = n \times \pi_i$$

$$df = k - 1$$

where  $k$  is the number of categories.

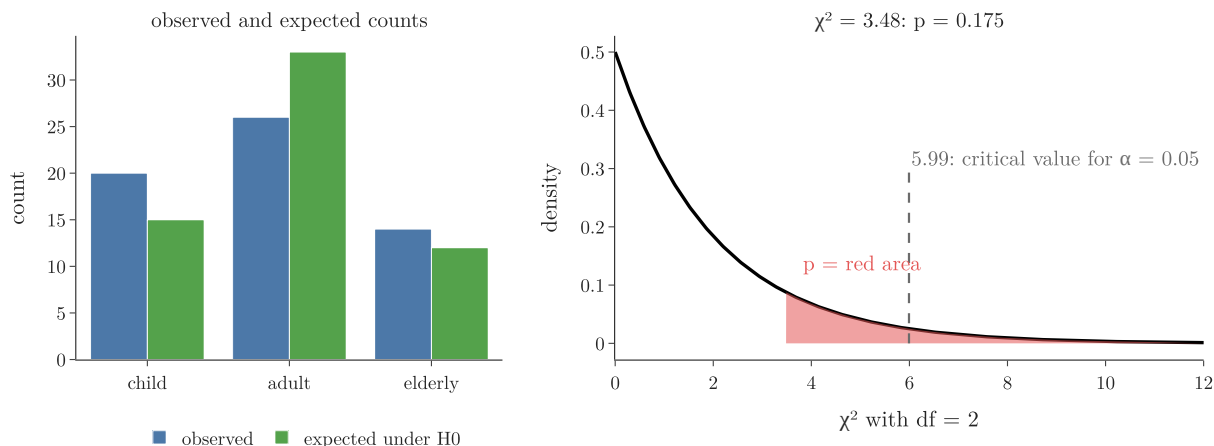
3. **Example:**

|                      | child | adult | elderly | total           |
|----------------------|-------|-------|---------|-----------------|
| observed $O$         | 20    | 26    | 14      | 60              |
| claimed share $\pi$  | 0.25  | 0.55  | 0.20    | 1               |
| expected $E = 60\pi$ | 15    | 33    | 12      | 60              |
| $(O - E)^2/E$        | 1.667 | 1.485 | 0.333   | $\chi^2 = 3.48$ |

The degrees of freedom are  $k - 1 = 2$ : once two expected counts are known, the third is fixed by the total of 60.

## 4.3 The decision

**Key point:**  $\chi^2 = 3.48$  with 2 df gives  $p = 0.175 > 0.05$ : we fail to reject  $H_0$ .



**Figure 5:** Goodness of fit for the age groups: the counts (left) and the p-value as a tail area (right)

Figure 5 shows both halves of the test. On the left, more children and fewer adults were observed than expected. On the right,  $\chi^2 = 3.48$  sits below the critical value 5.99, and the red tail area beyond it is  $p = 0.175$ .

Since  $0.175 > 0.05$ , we fail to reject  $H_0$ : the sample is consistent with the census proportions. The gaps on the left are within what chance produces in 60 people.

**Python:** `chisquare` takes the observed and expected counts.

```
from scipy import stats
```

```
stats.chisquare(f_obs=[20, 26, 14], f_exp=[15, 33, 12])
```

```
# statistic = 3.48, pvalue = 0.175
```

Without `f_exp`, `scipy` assumes equal proportions in all categories.

#### 4.4 Two categories: the proportion test again

**Key point:** With two categories, the goodness-of-fit  $\chi^2$  equals  $z^2$  of the one-sample proportion test, and the p-values are identical.

For gender (26 men, 34 women) against a 50/50 claim, the expected counts are 30 and 30:

$$\frac{(26 - 30)^2}{30} = 0.533$$

$$\frac{(34 - 30)^2}{30} = 0.533$$

$$\chi^2 = 0.533 + 0.533 = 1.067$$

With 1 df this gives  $p = 0.302$ . The proportion test in the [choosing a hypothesis test Note](#) gave  $z = -1.033$ , and its square is the same number:

$$(-1.033)^2 = 1.067$$

The p-value is the same too, 0.302. The goodness-of-fit test is the proportion test extended to any number of categories.

**Extra:** Goodness of fit also checks whether counts follow a named distribution, such as Poisson (see the [Poisson distribution Note](#)). The expected counts then come from the PMF, and every parameter estimated from the data (such as  $\lambda$  from the mean) removes one more degree of freedom:

$$df = k - 1 - (\text{number of estimated parameters})$$

The notebook tests this: 5,000 samples of 200 Poisson counts in 5 categories, with  $\lambda$  estimated each time, give a mean  $\chi^2$  of 3.03, matching  $df = 3$  rather than 4. Judged with the  $df = 4$  critical value, only 2.2% of the samples are rejected instead of 5%; with  $df = 3$  the rate is 5.1%.

## 5. The test of independence

**Key point:** Two categorical features in a contingency table: if they are independent, each expected count is row total  $\times$  column total / grand total, and  $df = (r - 1)(c - 1)$ .

### 5.1 The question and the table

**Key point:** Is the gender mix the same in every age group, or do gender and age group go together?

The **chi-square test of independence** (G-380) asks whether two categorical features are related. The counts go in a **contingency table** (G-464; see the [contingency tables Note](#)):

|              | child | adult | elderly | row total |
|--------------|-------|-------|---------|-----------|
| female       | 12    | 12    | 10      | 34        |
| male         | 8     | 14    | 4       | 26        |
| column total | 20    | 26    | 14      | 60        |

- $H_0$ : gender and age group are independent;
- $H_1$ : they are related.

## 5.2 Expected counts under independence

**Key point:** Independence means  $P(A \text{ and } B) = P(A)P(B)$ ; multiplied by  $n$ , this gives  $E = \text{row total} \times \text{column total}/n$ .

Start from the plain idea. Of the 60 people, 34 are female: a share of 56.7 percent:

$$\frac{34}{60} = 0.567$$

If age group has nothing to do with gender, every age group should have that same share of women. So of the 20 children we expect this many girls:

$$20 \times 0.567 = 11.33$$

And of the 26 adults we expect this many women:

$$26 \times 0.567 = 14.73$$

The same rule in the language of probability: two events are independent when the probability of both is the product of their probabilities (see the [independent events Note](#)). The **marginal probabilities** (G-1165) come from the totals (see the [joint and marginal probability Note](#)).

1. **In words:** the expected count of a cell is its row total times its column total, divided by the grand total.
2. **Formula:**

$$E = n \times \frac{\text{row total}}{n} \times \frac{\text{column total}}{n}$$
$$E = \frac{\text{row total} \times \text{column total}}{n}$$

3. **Example:** female children. The two probabilities are:

$$P(\text{female}) = \frac{34}{60}$$

$$P(\text{child}) = \frac{20}{60}$$

So the expected count is:

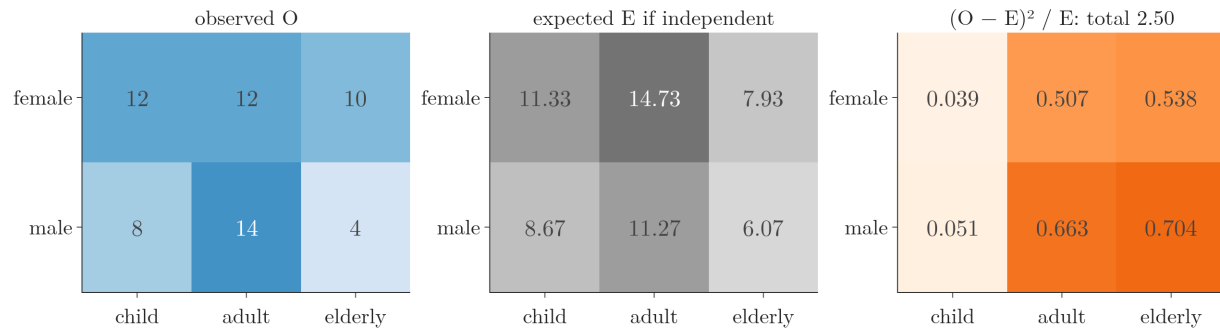
$$E = \frac{34 \times 20}{60}$$
$$E = \frac{680}{60} = 11.33$$

All six expected counts:

|              | child | adult | elderly | row total |
|--------------|-------|-------|---------|-----------|
| female       | 11.33 | 14.73 | 7.93    | 34        |
| male         | 8.67  | 11.27 | 6.07    | 26        |
| column total | 20    | 26    | 14      | 60        |

The expected table has the same totals as the observed one; only the inside is spread out “as if independent”.

Figure 6 sets the observed and expected tables side by side, with the third table holding each cell’s contribution to  $\chi^2$ , computed in the next section. The darkest contribution cells are the elderly men (4 observed, 6.07 expected) and the adult men (14 observed, 11.27 expected).



**Figure 6:** The test of independence for gender by age group. Left: observed counts. Middle: counts expected if gender and age group were independent, row total  $\times$  column total / 60. Right: each cell’s  $(O - E)^2 / E$ ; the six add up to  $\chi^2 = 2.50$ .

### 5.3 The statistic and the decision

**Key point:**  $\chi^2 = 2.50$  with 2 df gives  $p = 0.29$ : no evidence that gender and age group are related.

The contribution  $(O - E)^2 / E$  of each cell; for female children it is:

$$\frac{(12 - 11.33)^2}{11.33} = 0.039$$

All six:

|        | child | adult | elderly |
|--------|-------|-------|---------|
| female | 0.039 | 0.507 | 0.538   |
| male   | 0.051 | 0.663 | 0.704   |

1. **In words:** add the six contributions; the degrees of freedom are (rows  $- 1$ ) times (columns  $- 1$ ).

## 2. Formula:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$
$$df = (r - 1)(c - 1)$$

## 3. Example:

$$\begin{aligned}\chi^2 &= 0.039 + 0.507 + 0.538 \\ &\quad + 0.051 + 0.663 + 0.704 \\ \chi^2 &= 2.50 \\ df &= (2 - 1)(3 - 1) = 2\end{aligned}$$

With 2 df, the p-value is:

$$p = P(\chi^2 \geq 2.50) = 0.29$$

Since  $0.29 > 0.05$ , we fail to reject  $H_0$ . The elderly group has relatively few men (4 observed, 6.07 expected), but in a sample of 60 a gap like this is common by chance.

Why  $(r - 1)(c - 1)$ : with the row and column totals fixed, filling  $(r - 1)(c - 1)$  cells freely determines all the others. In a 2 by 3 table, choosing 2 cells of the female row fixes the rest.

**Python:** `chi2_contingency` takes the observed table and returns the statistic, p-value, df and expected counts.

```
import pandas as pd
```

```
table = pd.crosstab(people["gender"], people["age_group"])
result = stats.chi2_contingency(table)
result.statistic, result.pvalue, result.dof      # 2.50, 0.29, 2
result.expected_freq                             # the expected table
```

## 6. Case study: survival on the Titanic

**Key point:** Sex and class were both strongly related to survival:  $\chi^2 = 263$  ( $p \approx 10^{-59}$ ) for sex,  $\chi^2 = 103$  ( $p \approx 10^{-23}$ ) for class.

### 6.1 Sex and survival

**Key point:** 81 women died where independence predicts 194; 109 men survived where it predicts 222.

The Kaggle Titanic training file has 891 passengers. Survival against sex:

|                  | died | survived |
|------------------|------|----------|
| female, observed | 81   | 233      |

|                  | died  | survived |
|------------------|-------|----------|
| female, expected | 193.5 | 120.5    |
| male, observed   | 468   | 109      |
| male, expected   | 355.5 | 221.5    |

Each cell contributes (observed minus expected) squared, divided by expected:

| Cell             | Work                      | Contribution |
|------------------|---------------------------|--------------|
| female, died     | $(81 - 193.5)^2 / 193.5$  | 65.4         |
| female, survived | $(233 - 120.5)^2 / 120.5$ | 105.0        |
| male, died       | $(468 - 355.5)^2 / 355.5$ | 35.6         |
| male, survived   | $(109 - 221.5)^2 / 221.5$ | 57.1         |

The four contributions add up to  $\chi^2 = 263.1$  with  $df = 1$ , and the p-value is tiny:

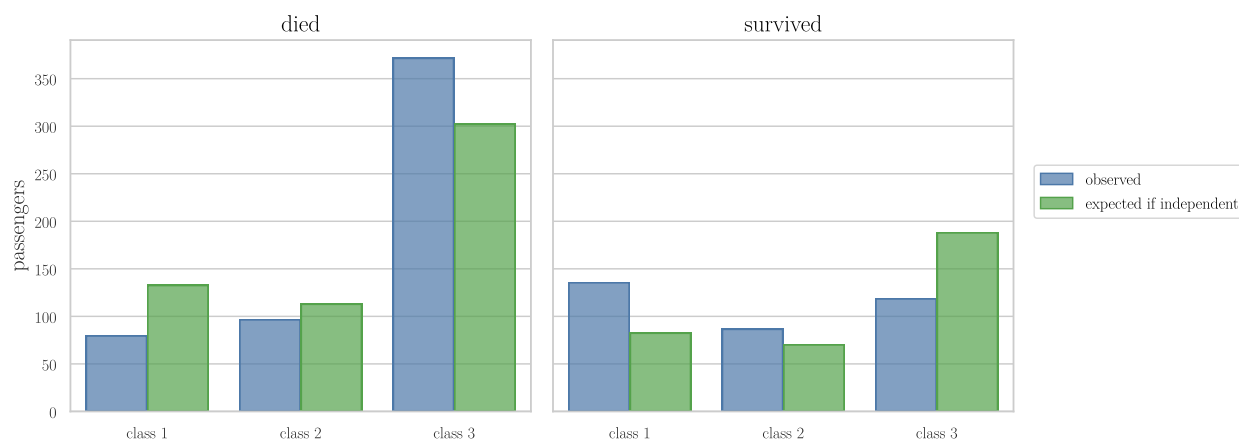
$$p = 3.7 \times 10^{-59}$$

We reject  $H_0$ : survival depended on sex.

**Extra:** For a 2 by 2 table, `chi2_contingency` applies **Yates' continuity correction** (G-2135) by default: it shrinks each  $|O - E|$  by 0.5 before squaring (Yates 1934), which lowers  $\chi^2$  and so makes the test slightly more cautious. Here it gives  $\chi^2 = 260.7$  instead of 263.1; pass `correction=False` to get the plain statistic. With counts this large the choice does not matter.

## 6.2 Class and survival

**Key point:** First class had 136 survivors where independence predicts 83; third class had 119 where it predicts 188.



**Figure 7:** Titanic passengers by class and outcome: observed counts and counts expected under independence

The crosstab of class against survival was drawn as a heatmap in the [bivariate analysis Note](#). Figure 7 compares its observed counts with the expected ones. First class has far more survivors than independence predicts, third class far fewer. The survival rates were 63%, 47% and 24% for the three classes, against 38% overall.

The test gives  $\chi^2 = 102.9$  with 2 degrees of freedom:

$$df = (3 - 1)(2 - 1) = 2$$

The p-value is tiny:

$$p = 4.5 \times 10^{-23}$$

We reject  $H_0$ : survival depended on class.

## 7. Conditions and limits

**Key point:** Use raw counts, independent observations, and expected counts of at least 5 in every cell; a significant  $\chi^2$  says that a relationship exists, not how strong it is.

### 7.1 Conditions

**Key point:** Counts, independence, and large enough expected counts.

1. **Counts, not percentages.** The formula needs the number of observations in each cell. Percentages or means give a wrong  $\chi^2$ .
2. **Independent observations.** Each observation is counted in exactly one cell, and observations do not influence each other. The same person measured twice breaks this.

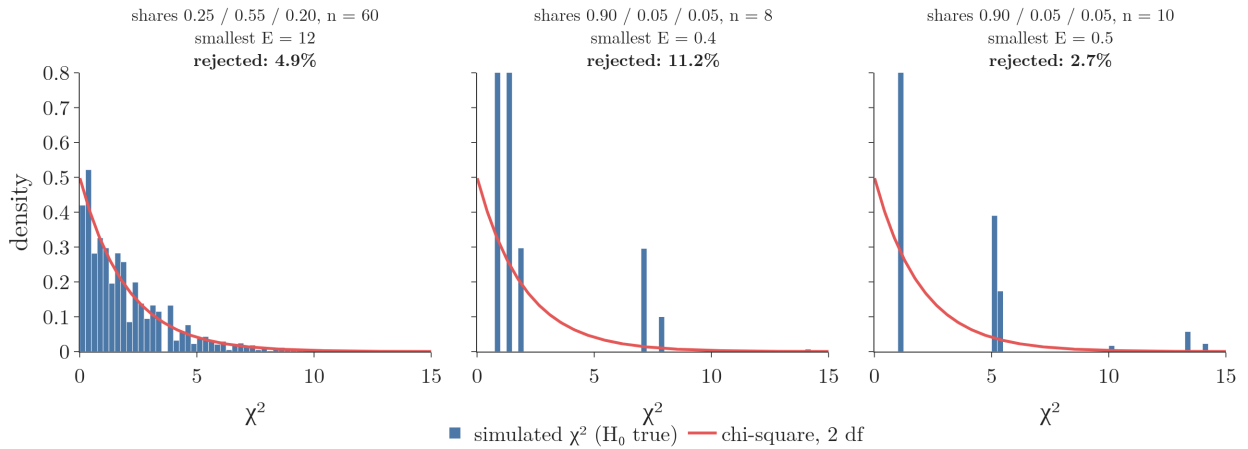
3. **Large enough expected counts.** The usual rule is that every expected count should be at least 5 (Cochran 1954), because the chi-square distribution is only a large-sample approximation. All six expected counts of section 5.2 are above 5 (the smallest is 6.07). For small 2 by 2 tables, **Fisher's exact test** (G-782) (`stats.fisher_exact`) avoids the approximation; for bigger tables, merging rare categories helps.

**Extra:** What goes wrong below the rule. A ruler marked in metres is fine for a room but useless for a grain of rice; the chi-square curve is likewise only a good ruler for large counts. The notebook draws 200,000 samples with  $H_0$  true, so a correct 5% test should reject 5% of them:

| Claimed shares   | $n$ | Smallest $E$ | Rejected at 5% |
|------------------|-----|--------------|----------------|
| 0.25, 0.55, 0.20 | 60  | 12           | 4.9%           |
| 0.90, 0.05, 0.05 | 8   | 0.4          | 11.2%          |
| 0.90, 0.05, 0.05 | 10  | 0.5          | 2.7%           |

With large expected counts the test keeps its 5% promise. With two rare categories and tiny samples, the false-alarm rate drifts away from 5%, more than double in one case and about half in the other, so the p-value can no longer be trusted.

Figure 8 shows why. With large expected counts the simulated  $\chi^2$  values spread smoothly along the chi-square curve. With expected counts below 1, only a few count patterns are possible, so  $\chi^2$  can take only a few values: it piles up on a handful of spikes that the smooth curve cannot describe, and the share beyond the 5% line (dashed) is wrong.



**Figure 8:** Simulated  $\chi^2$  values with  $H_0$  true, 200,000 samples each (bars), against the chi-square distribution with 2 df (curve). Left: every expected count is at least 12, and the curve fits. Middle and right: expected counts of 0.4 and 0.5 make  $\chi^2$  pile up on a few values, and the share beyond the 5% critical value (dashed) becomes 11.2% or 2.7%.

## 7.2 Strength of a relationship: Cramér's V

**Key point: Cramér's V** (G-500) rescales  $\chi^2$  to a number between 0 (no relationship) and 1 (perfect relationship) (Cramér 1946).

$\chi^2$  grows with the sample size: the same pattern in 10 times as many observations gives a 10 times larger  $\chi^2$ . To measure strength, we divide the size out.

1. **In words:** divide  $\chi^2$  by the sample size and by one less than the smaller table dimension, then take the square root.
2. **Formula:**

$$V = \sqrt{\frac{\chi^2}{n \times (\min(r, c) - 1)}}$$

3. **Example:** Titanic sex and survival,  $\chi^2 = 263.1$ ,  $n = 891$ , a 2 by 2 table:

$$\frac{263.1}{891 \times 1} = 0.295$$

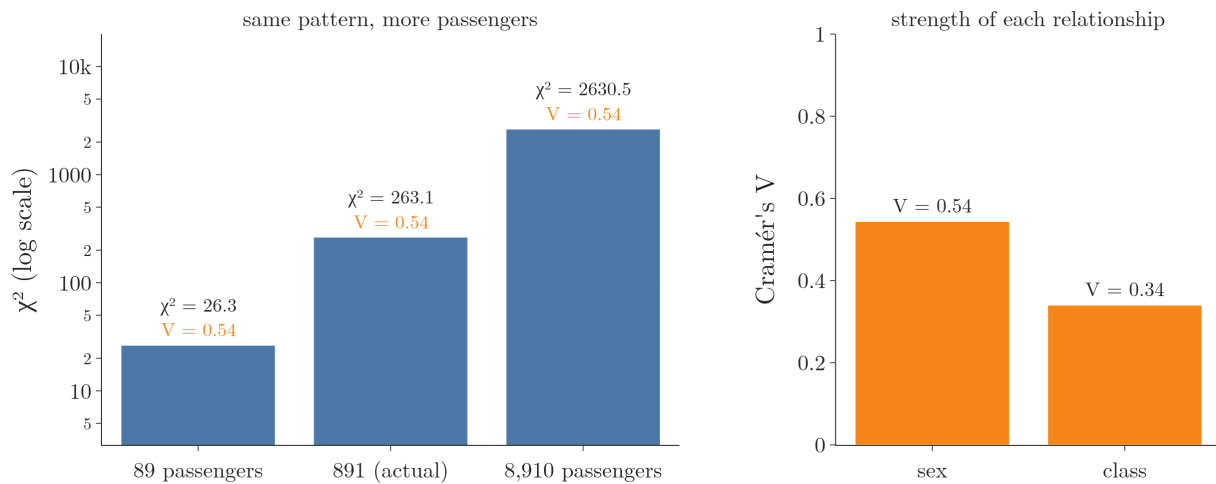
$$V = \sqrt{0.295} = 0.54$$

For class and survival:

$$V = \sqrt{\frac{102.9}{891}} = 0.34$$

Both relationships are significant, but sex was the stronger predictor of survival. In scipy: `stats.contingency.association(table, method="cramer")`.

Figure 9 shows why the rescaling is needed. The Titanic sex-by-survival table is shrunk to a tenth and grown to ten times its size, keeping the same pattern.  $\chi^2$  grows with the number of passengers, from 26.3 to 263.1 to 2,630.5, but  $V$  stays at 0.54 every time.



**Figure 9:** Left: the Titanic sex-by-survival pattern with a tenth, the actual and ten times the passengers.  $\chi^2$  (blue, log scale) grows with the sample; Cramér's V stays 0.54. Right: V for sex (0.54) and for class (0.34) against survival.

### 7.3 Chi-square in machine learning

**Key point:** Feature selection with `SelectKBest(chi2)` ranks categorical features by how strongly they are related to the target.

**Feature selection** (G-768) with **SelectKBest** (G-1762) (`score_func=chi2`) (see the [pipelines Note](#)) scores each feature against the **target** (G-1949; the output we predict) with a chi-square statistic: a feature whose counts differ strongly between the classes gets a high score. scikit-learn's version treats each feature's values as counts, which is why it needs values of 0 or more (scikit-learn docs, `chi2`). The score is not the same number as the test of independence. For a one-hot column it compares only the rows where the column is 1. In the notebook, Titanic sex one-hot encoded gets the scores 170.3 (female) and 92.7 (male); only their sum, 263.1, equals the  $\chi^2$  of section 6.1.

## 8. Summary

|                    | Goodness of fit                                     | Test of independence                              |
|--------------------|-----------------------------------------------------|---------------------------------------------------|
| Features           | one categorical                                     | two categorical                                   |
| $H_0$              | the categories have the claimed proportions         | the two features are independent                  |
| Expected count     | $n \times \pi_i$                                    | row total $\times$ column total $/n$              |
| Degrees of freedom | $k - 1$                                             | $(r - 1)(c - 1)$                                  |
| scipy              | <code>chisquare</code>                              | <code>chi2_contingency</code>                     |
| Example here       | age groups vs census:<br>$\chi^2 = 3.48, p = 0.175$ | gender vs age group:<br>$\chi^2 = 2.50, p = 0.29$ |

- Under  $H_0$ ,  $\chi^2 = \sum(O - E)^2/E$  follows a chi-square distribution; the p-value is the right-tail area.
- With two categories, goodness of fit is the one-sample proportion test:  $\chi^2 = z^2$ .
- The test needs raw counts, independent observations and expected counts of at least 5.

## 9. Sources

### Built from

- Krish Naik, “Tutorial 32- All About P Value,T test,Chi Square Test, Anova Test and When to Use What?”, YouTube, <https://www.youtube.com/watch?v=YrhlQB3mQFI>
- Khan Academy, “Chi-square distribution introduction | Probability and Statistics | Khan Academy”, YouTube, <https://www.youtube.com/watch?v=dXB3cUGnaxQ>
- Khan Academy, “Pearson’s chi square test (goodness of fit) | Probability and Statistics | Khan Academy”, YouTube, <https://www.youtube.com/watch?v=2QeDRsxSF9M>
- Khan Academy, “Contingency table chi-square test | Probability and Statistics | Khan Academy”, YouTube, <https://www.youtube.com/watch?v=hpWdDmgsIRE>

### Other references

- Cochran, W. G. (1954). “Some Methods for Strengthening the Common  $\chi^2$  Tests”. *Biometrics* 10(4).
- Cramér, H. (1946). *Mathematical Methods of Statistics*. Princeton University Press.
- NIST/SEMATECH (2012). *e-Handbook of Statistical Methods*. [itl.nist.gov/div898/handbook](http://itl.nist.gov/div898/handbook). Section 1.3.6.6.6, chi-square distribution.
- Pearson, K. (1900). “On the criterion that a given system of deviations from the probable in the case of a correlated system of variables is such that it can be reasonably supposed to have arisen from random sampling”. *Philosophical Magazine* 50(302).
- scikit-learn documentation. `sklearn.feature_selection.chi2`.
- Yates, F. (1934). “Contingency Tables Involving Small Numbers and the  $\chi^2$  Test”. *Supplement to the Journal of the Royal Statistical Society* 1(2).

## 10. Key terms

| Term                            | Meaning                                                                                                       |
|---------------------------------|---------------------------------------------------------------------------------------------------------------|
| Observed count $O$              | The number of observations in a category or table cell                                                        |
| Expected count $E$              | The number of observations a category or cell would hold on average if $H_0$ were true                        |
| Chi-square statistic $\chi^2$   | $\sum(O - E)^2/E$ : the total mismatch between observed and expected counts                                   |
| Chi-square distribution         | The distribution of $\chi^2$ under $H_0$ : positive, right-skewed, with one parameter, the degrees of freedom |
| Goodness-of-fit test            | A chi-square test of whether one categorical feature follows claimed proportions;<br>$df = k - 1$             |
| Chi-square test of independence | A chi-square test of whether two categorical features are related; $df = (r - 1)(c - 1)$                      |
| Yates' continuity correction    | A small adjustment to $\chi^2$ for 2 by 2 tables, applied by default in <code>chi2_contingency</code>         |
| Fisher's exact test             | An exact test for small 2 by 2 tables, used when expected counts fall below 5                                 |